

Suvam Basak

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Prime Minister's Research Fellow at *Indian Institute of Technology Kanpur*

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Research Interests

Implementation and simulation of *Networked Systems* and the *Internet of Things*.

Education

Ph.D.	Indian Institute of Technology Kanpur (IIT-K) Department of Computer Science & Engineering Advisor: Dr. Amitangshu Pal & Dr. Debopam Bhattacharjee – CGPA: 8.75/10	Kanpur, Uttar Pradesh 2022 – Present
M.Tech.	University of Hyderabad (UoH) School of Computer and Information Sciences collaboration with IDRBT Thesis title: An Integrated and Standards-based Fog Computing Federation Framework Advisor: Prof. Satish Narayana Srirama – CGPA: 9.61/10	Hyderabad, Telangana 2020 – 2022
B.Tech.	West Bengal University of Technology (WBUT) Computer Science & Engineering, JIS College of Engineering CGPA: 8.6/10	Kolkata, West Bengal 2014 – 2018

Conference Publications

2025 USENIX ATC	LEOCraft: Towards Designing Performant LEO Networks – S. Basak, A. Pal, D. Bhattacharjee
2024 ACM IMC	CosmicDance: Measuring Low Earth Orbital Shifts due to Solar Radiations – S. Basak, A. Pal, D. Bhattacharjee
2022 IEEE Cloud	Fog Computing out of the Box with FogDEFT Framework: A Case Study – S. N. Srirama, S. Basak

Journal Publications

2023 ACM IJNM	Fog computing out of the box: Dynamic deployment of fog service containers with TOSCA – S. Basak, S. N. Srirama
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Workshop Publications

2025 SIGCOMM LEO-NET	An investigation of Starlink's performance during the May '24 solar superstorm – S. Basak, A. Pal, D. Bhattacharjee
2023 MobiCom LEO-NET	Exploring Low-Earth Orbit Network Design – S. Basak, A. Pal, D. Bhattacharjee

Talks

Aug 9, 2025 – IIT Kanpur	LEOCraft: Towards Designing Performant LEO Networks – CSE Research Day
Jul 8, 2025 – Boston, USA	LEOCraft: Towards Designing Performant LEO Networks – USENIX ATC
Apr 6, 2025 – IIT Kanpur	Measuring Orbital Shifts Due to Solar Radiation – CS801 Research Colloquium
Nov 4, 2024 – Madrid, Spain	CosmicDance: Measuring Low Earth Orbital Shifts due to Solar Radiation – ACM IMC

Research Experience

Cloud and Smart Labs Advisor: Prof. Satish N. Srirama (UoH)	2021 – 2022
An Integrated and Standards-based Fog Computing Federation Framework: Dynamic deployment of services on the fog nodes out of the box on the fly with OASIS TOSCA standard. GitHub .	

Teaching Experience

UEM Kolkata

CSJM University (UIET)

IIT Kanpur

Course Instructor

- Advanced Python Programming 2024-25

- CSE-S208 - Data Structure Using Python 2023-24-II

Teaching Assistant, Department of CSE

- [ESC111/112](#) - Fundamentals of Computing - I/II 2025-26-I

- [ESC111/112](#) - Fundamentals of Computing - I/II 2024-25-II

- [CS724](#) - Sensing, Communications and Networking for Smart Wireless Devices 2023-24-I

Teaching Assistant, eMasters in Financial Technology & Management

- [MBA977](#) - IT Platforms, Cloud Computing, and IoT 2023-24

Teaching Assistant, eMasters in Cyber Security

- [CS971](#) - Computer Networking II 2022-23

- [CS963](#) - Computer Networking I 2022-23

Tutor, Department of CSE

- [ESC111/112](#) - Fundamentals of Computing - I/II 2022-23-II

- [ESC111/112](#) - Fundamentals of Computing - I/II 2022-23-I

Honors and Scholarships

Prime Minister's Research Fellowship (PMRF) — Cycle 11 2023 – 2026

Smart India Hackathon finalist (KL University, Guntur, Andhra Pradesh) 2017

Technical Skills

Programming:

Tools & Softwares:

Hardware:

Proficient in: C, Python — Familiar with: Flask, Verilog, Dart, Flutter, Solidity, JavaScript, Java

Gurobi Optimizer, Ansible, Docker, Docker Swarm, TOSCA, OpenStack, QEMU/KVM

Raspberry Pi, Arduino

Referees

IIT Kanpur

Dr. Amitangshu Pal, Assistant Professor

Department of Computer Science & Engineering

E-mail: amitangshu@cse.iitk.ac.in, Web: cse.iitk.ac.in/users/amitangshu

Microsoft Research

Dr. Debopam Bhattacharjee, Senior Researcher

E-mail: debopamb@microsoft.com, Web: bdebopam.github.io

University of Hyderabad

Prof. Satish Narayana Srirama, Professor (Member IEEE)

School of Computer and Information Sciences

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